MORROW 8:00 PM/ET
ROCKETS
PLAYOFFS
State Farm
POR LEADS SERIES 3 - 1
BONUS
BONUS
OKC 115
POR 115
4TH 05.3

WHAT DRIVES NBA OFFENSE MORE?

SHOT CREATION

SHOT MAKING

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HYPOTHESIS

Shot Creation has a greater association on offensive rating

BACKGROUND INFO

- NBA Seasons 2015-2025 (330 Teams)
- Success measure: Offensive Rating
- Independent Variables: Shot Creation and Shot Making
- Shot Creation: Opportunity before the ball is released
- Shot Making: Opportunity after the ball is released

NBA DATA TO MODEL

1) COLLECT

NBA StatsAPI & Shot Maps

(2015-2025)

2) ORGANIZE

One big Dataset with one row per Team Season(330)

3) ANALYZE METRICS

Shot Creation, Shot Making & Shooting Stats

4) TEST

Predict O-Rating using two Linear Regression Models(OLS)

SHOT CREATION

What quality of opportunities did the offense produce?(Before)

ASSOCIATION

SHOT MAKING

How well did the team convert its opportunities?(After)

SHOT CREATION AMD SHOT MAKING CONSTRUCTION

SHOT CREATION (ORIGINAL)

SHOT CREATION = Z(Rim Rate) + Z(Corner Three Rate) + Z(Wide Open Rate) – Z(Bad Shot Rate)

SHOT MAKING (ORIGINAL)

Shot Making = 100(Actual EFG - Expected EFG) in %pts

Expected eFG = Location Expected eFG + Defender Adjustment

SHOT CREATION (REVISED)

SHOT CREATION = Z(Rim Rate) + Z(Corner Three Rate) + Z(Wide Open Rate) – Z(Bad Shot Rate)

SHOT MAKING (REVISED)

Shot Making = 100(Actual EFG - Expected EFG) in %pts

Expected eFG = Location Expected eFG + Defender Adjustment

SHOT CREATION (FINAL)

SHOT CREATION = Z(Rim Rate) + Z(Corner Three Rate) + Z(Wide Open Rate) – Z(Bad Shot Rate)

SHOT MAKING (FINAL)

Shot Making = 100(Actual EFG - Expected EFG) in %pts

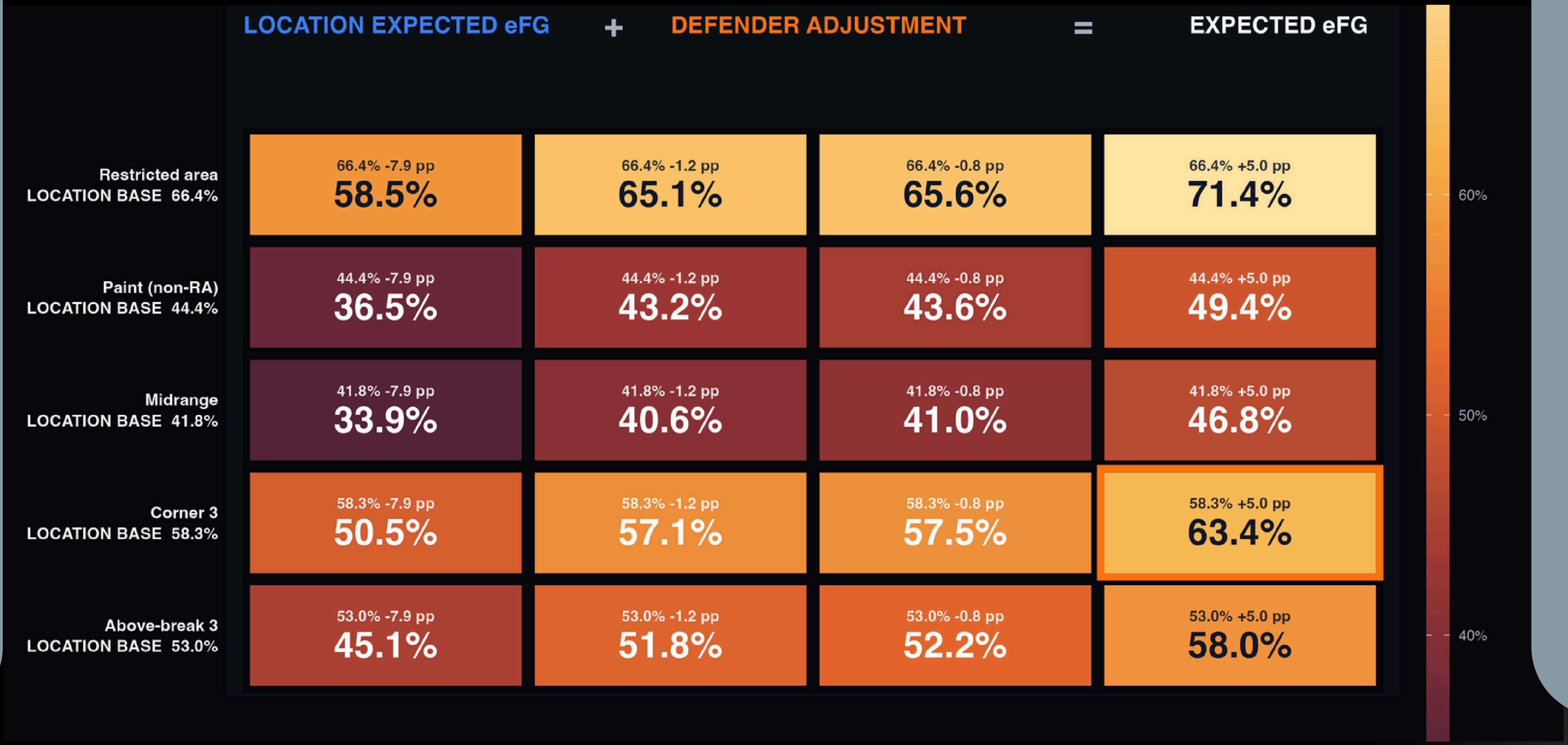
Expected eFG = Location Expected eFG + Defender Adjustment

SHOT MAKING CONSTRUCTION

STEP 1:

Expected eFG = Location Expected eFG + Defender Adjustment

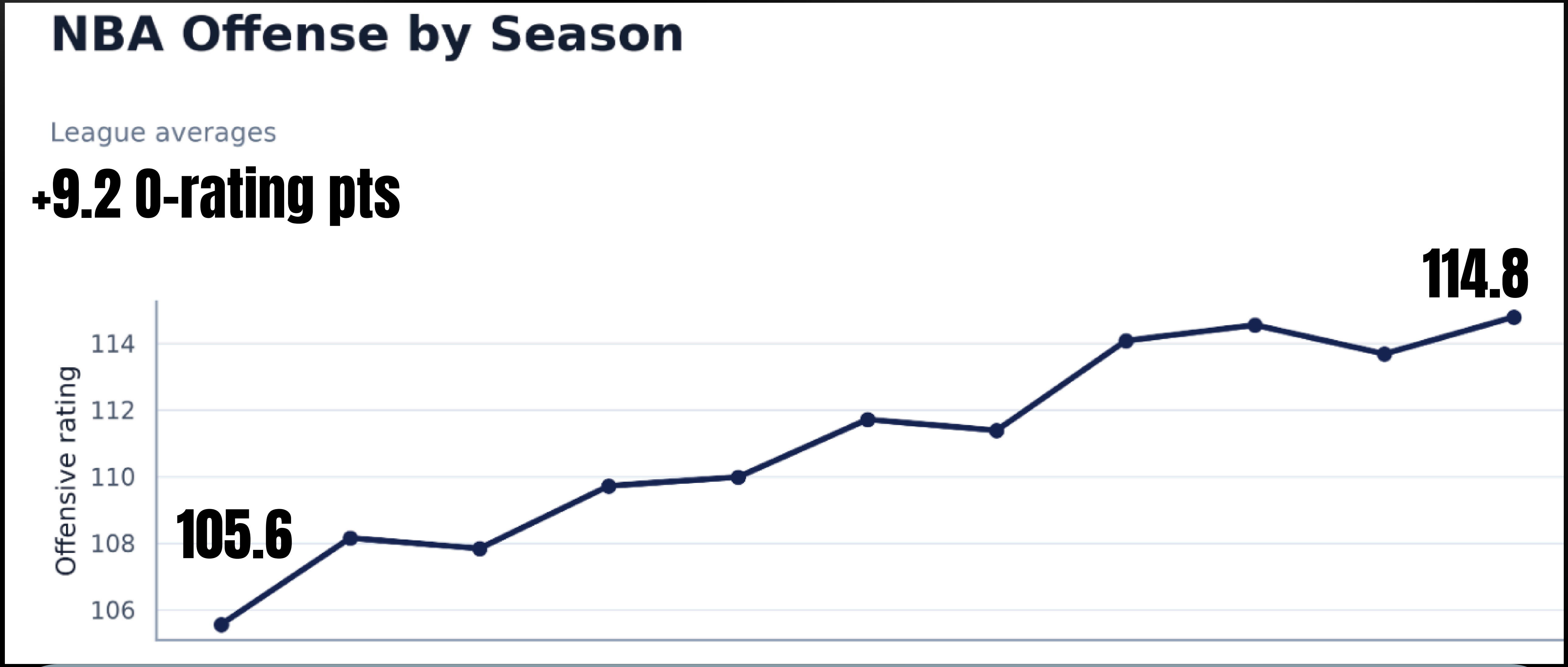
STEP 2:



STEP 3:

Shot Making = 100(Actual EFG - Expected EFG) in %pts

NBA OFFENSE CHANGED ACROSS ERAS



2015-2016 SEASON TO 2025-2026 SEASON

Season Fixed Effects

2016 <-> 2016

2025 <-> 2025

- Own ORtg baseline per season
- Teams compared in same season

REGRESSION ANALYSIS

Model 1:

$$\text{ORtg} = X_0 + X_1 Z(\text{Creation}) + X_2 Z(\text{Making}) + \text{Season} + \varepsilon$$

Question: Which shooting skill has the stronger within-season association with Offensive Rating?

Predictors: Shot Creation, Shot Making and season basline data.

Coefficients(x_n): represents strength between independent variable and O-rating

Model 2:

$$\text{ORtg} = X_0 + x_1 Z(\text{Creation}) + x_2 Z(\text{Making}) + x_3 Z(\text{TOV\%}) + x_4 Z(\text{ORB\%}) + x_5 Z(\text{FTr}) + \text{Season} + \varepsilon$$

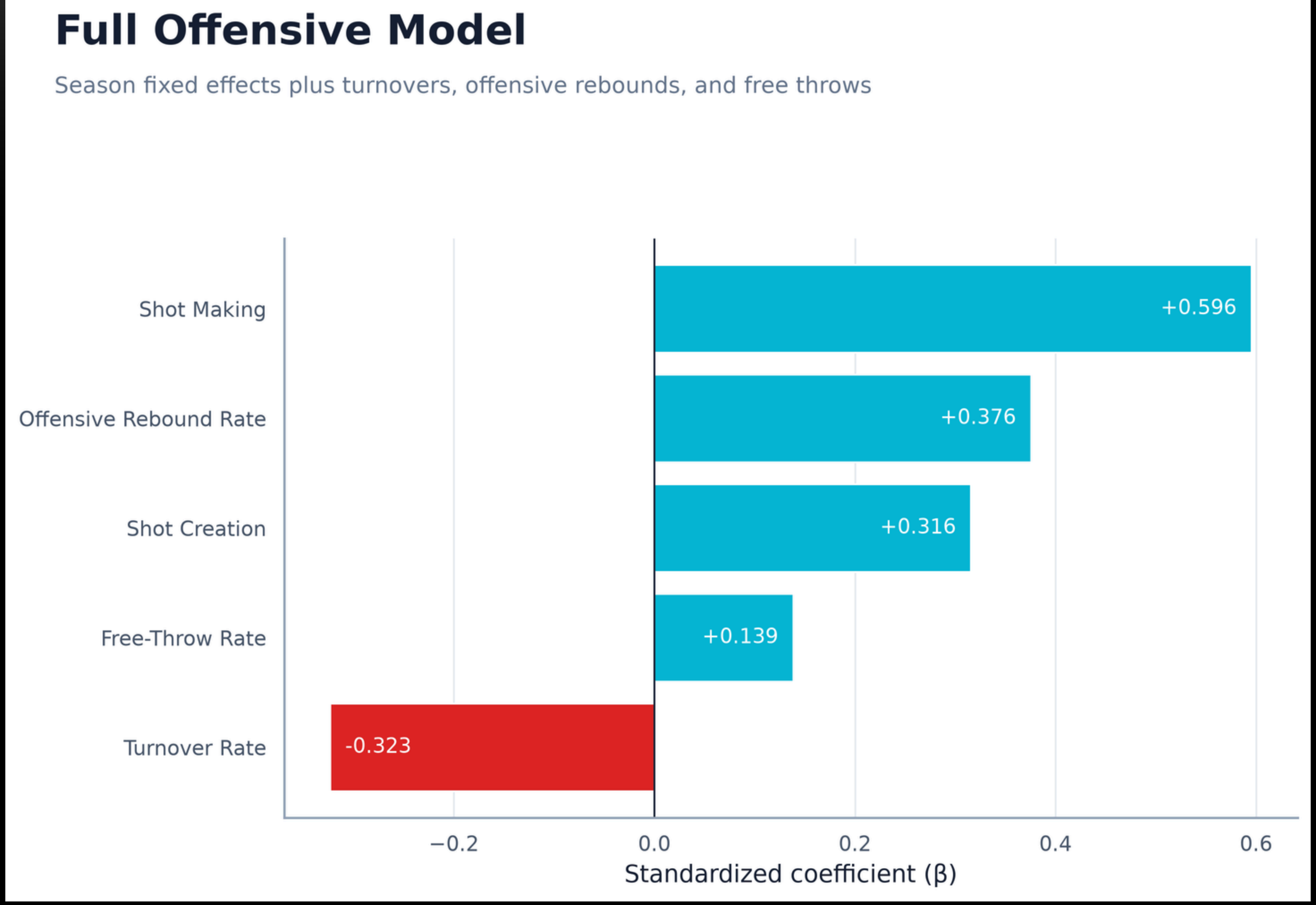
Question: Does the result remain after accounting for other major offensive factors(controll)?

Additional controls: Turnover rate, offensive rebound rate and free-throw rate.

X_0: The predicted ORtg for a 2015-16 team with average Shot Creation and average Shot Making.

*OLS model predicts intercept and coeffecients by minimizing the sum of the squared error of the residual of O-rating

SHOT MAKING REMAINS STRONGEST AFTER CONTROLS IN M2



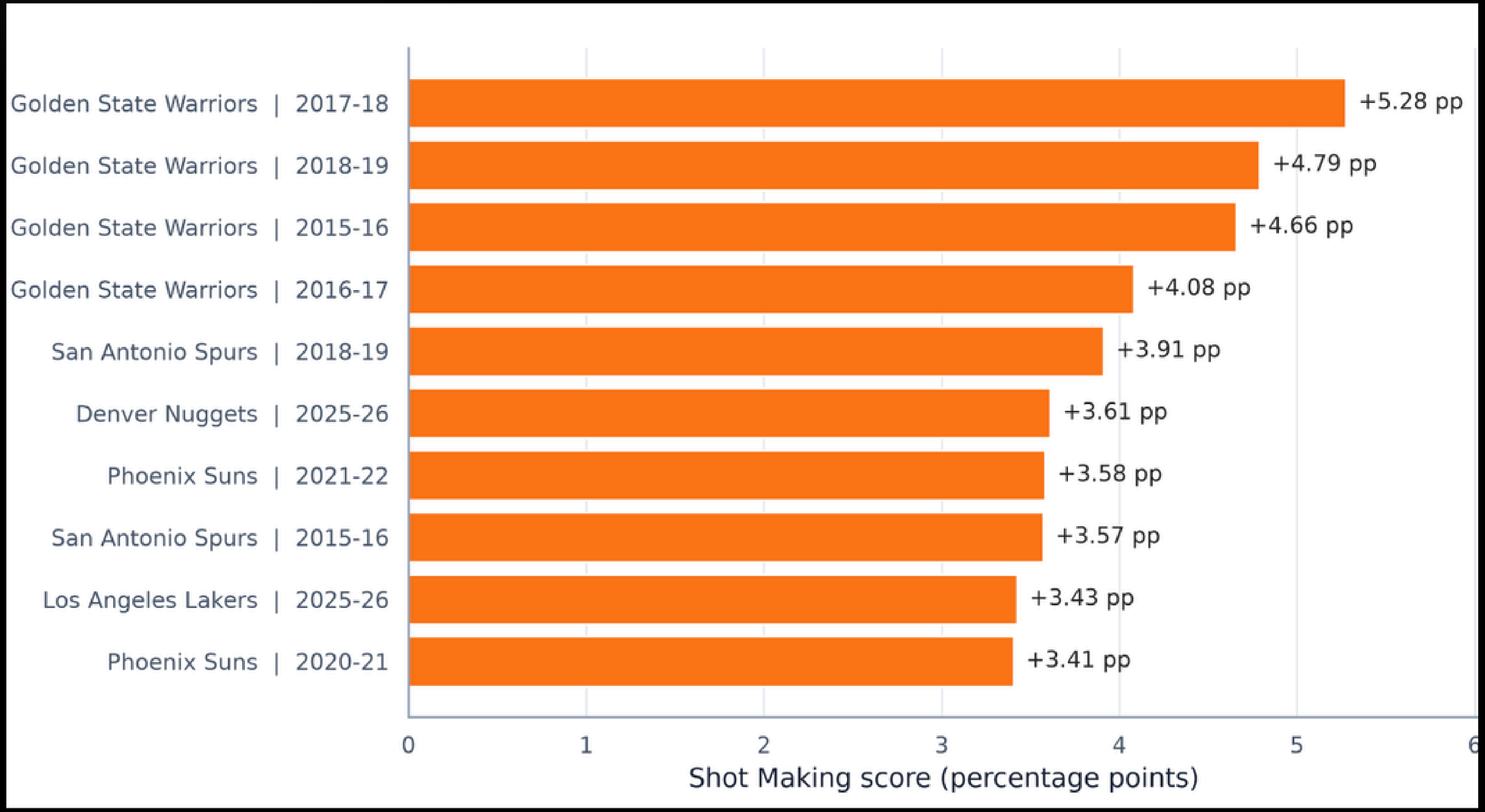
Shot Making: 0.597 to 0.596 after controlling

Shot Creation: 0.195 to 0.316 after controlling

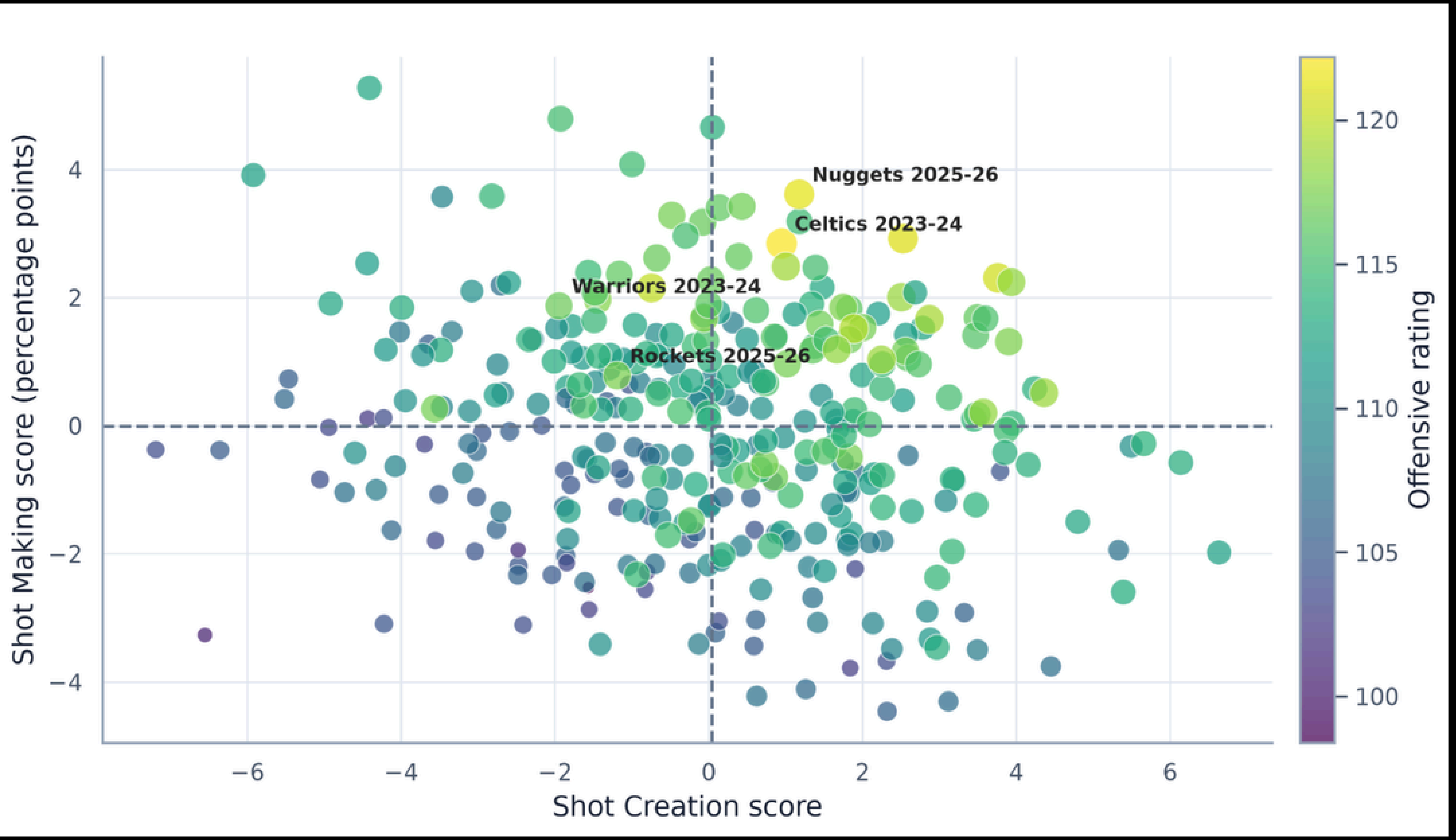
WHY?: M2 exposes control variables separately to shot creation while M1 mixed factors to create score

WHO BEAT OFFENSIVE EXPECTATIONS

Top Shot Making Team Seasons(%pts)



Team Shot Creation + Shot Making



Dashboard

EXPERT INTEREST

Sameer Deshpande

statistician whose research develops advanced Bayesian methods to analyze sports, with influential work in basketball, baseball, and football analytics.

Luke Bornn

Former Harvard Assistand Professor of Statistics, VP of Strategy and Analytics for the Kings. Specialzies in player tracking data and ML in soccer and basketball.

Tyler Skinner

Assistant Professor of Sport Management at the University of South Carolina. Specializes in sports analytics, sports economics, and leadership.

LIMITATIONS

1. Custom metric: Shot Creation is an equal-weight index, not an official NBA metric.
2. Aggregated data: location and defender distance are separate team-season tables, not shot-level data.
3. Mechanical relationship: Shot Making uses eFG%, which is directly connected to points per possession.
4. Observational design: coefficients show associations, not causal effects.
5. Unobserved team differences: season fixed effects included, team fixed effects are not.

THANK YOU FOR LISTENING

THE RESULT IS STABLE ACROSS MODEL CHOICES

Shot Making had the larger coefficient in 38 of 40 sensitivity specifications.

Location-only expectation: 19 of 20
Location + defender expectation: 19 of 20
All specifications: 38 of 40

38 OF 40

APPENDIX: COMPLETE FORMULA REFERENCE I

Shot Location & Defender Rates

- Rim Rate = Restricted-Area FGA / Total FGA
- Corner3 Rate = Corner3 FGA / Total FGA
- Midrange Rate = Midrange FGA / Total FGA
- Wide Open Rate = WideOpen FGA / Defender Total
- Contested Rate = (VeryTight FGA + Tight FGA) / Defender Total
- BadShot Rate = (Midrange Rate + Contested Rate) / 2

Z-score & Shot Creation Score

- $Z(x) = (x - \text{mean}(x)) / SD(x)$
- Shot Creation Score = Z(Rim Rate) + Z(Corner 3 Rate) + Z(Wide-Open Rate) − Z(Bad-Shot Rate)

Location Expected eFG%

- Location Expected eFG% = Σ (team zone frequency × season league-average zone eFG%)

League Zone eFG%

- League zone eFG%(Three-Point Zones) = League zone FG% × 1.5
- League zone eFG%(Two-Point Zones) = League zone FG%